Unified Table Understanding via Context-Aware Cell Embedding Learning

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***Abstract*—The rapid progress of Large Language Models (LLMs) has diverted attention from how table cells should be represented, the tabular analogue of word embeddings in NLP. Processing tables as flat token sequences, LLMs overlook cell-level semantics and lack a single, reusable representation that transfers across Table Understanding (TU) tasks. We introduce Unified Table Understanding via Embedding Learning (UTUEL), a self-supervised, task-agnostic framework that casts Table Representation Learning as a Joint-Embedding Predictive Architecture (TRL-JEPA). Each cell is linearized with its row and column context through a Semantic Meta-Path (SMP); an encoder refines the initial LLM embeddings into context-aware cell-level embeddings, and a predictor recovers a cell's context embedding from its latent representation, distinguishing identical values in different contexts. Learned once per table, these embeddings are reused across queries and pooled to column- or table-level for different TU tasks. On a new benchmark and a large-scale dataset, UTUEL supports online Table Question Answering and, reusing the same embeddings without retraining, substantially improves Table Retrieval over generic embedders while remaining competitive on Column Type Annotation and Table Cell Retrieval. It achieves this with a compact 2.9–11.8M-parameter encoder, orders of magnitude smaller than the evaluated baselines in both parameters and per-query token cost.**

***Index Terms*—Table understanding, table representation learning, joint-embedding predictive architecture, self-supervised learning, table embedder.**

I. Introduction

TABLE Understanding (TU), which extracts insights from structured data (databases, spreadsheets) and semi-structured data (web tables, document-embedded tables), faces three challenges that no existing framework resolves jointly. First, tables exhibit *semantic complexity*, where a cell has no well-defined meaning in isolation and, unlike a word in natural language, derives its meaning only from the table in which it appears. Second, they exhibit *contextual variability:* the same value can carry different meanings depending on its headers; for example, `12` may denote age under *Person* × *Age* but tenure under *Person* × *Years of experience*. Third, they exhibit *structural heterogeneity:* tables vary widely in layout, including hierarchical headers, merged cells, and transposed structures.

Recent TU frameworks for Table Retrieval (TR) [1], Table Question Answering (TQA) [2], [3], [4], [5], Column Type Annotation (CTA) [6], and related tasks [7], [8] address these challenges task by task, rather than through a task-agnostic *cell-level embedding* that could expose shared properties such as permutation invariance [9], [10]. This contrasts with NLP, where a single word embedding is learned once and reused across applications such as recommendation and Retrieval-Augmented Generation (RAG). A table, however, is a *self-contained modality:* each cell's meaning is informed by its own rows and headers, henceforth its *context*. Exploiting this, we propose UTUEL (Unified TU via context-aware cell Embedding Learning), which treats a table's relational structure as its own source of supervision, without explicit queries or task labels; its workflow is illustrated in Fig. 1.

UTUEL constructs cell embeddings by linearizing each cell with its row and column headers via the Semantic Meta-Path (SMP) [11], such that cells with the same surface value receive different representations depending on their context. This jointly addresses the three TU challenges: a well-defined cell semantics within each SMP, addressing the semantic complexity; a distinct SMP per header combination, capturing the contextual variability; and a layout-preserving linearization, disentangling the structural heterogeneity. As a table-domain counterpart to word embeddings, this unified representation supports multiple tasks: cell-level embeddings enable Table Cell Retrieval (TCR), and thus TQA; averaged cell embeddings provide column representations for CTA; and aggregating all cell embeddings yields table-level representations for TR.

Because UTUEL refines rather than creates an embedding space, it requires an LLM embedder provider for the initial embedding, which is both its natural source and a demanding baseline: recent benchmarks [12], [13], [14], [15], [16] report that general-purpose LLMs such as GPT-OSS [17] and Qwen3.6 [18] match or exceed table-specialized models such as TableGPT2 [14], [19] and GritLM [20], but at substantial inference cost. Rather than deploy them end-to-end, UTUEL applies a lightweight encoder over HuggingFace and Ollama-served LLM embedders (22M–0.6B parameters), so its performance derives from contextualization rather than scale. We evaluate UTUEL across several datasets and the above TU tasks against both

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Code and data: https://github.com/DIMA-VUB/UTUEL.

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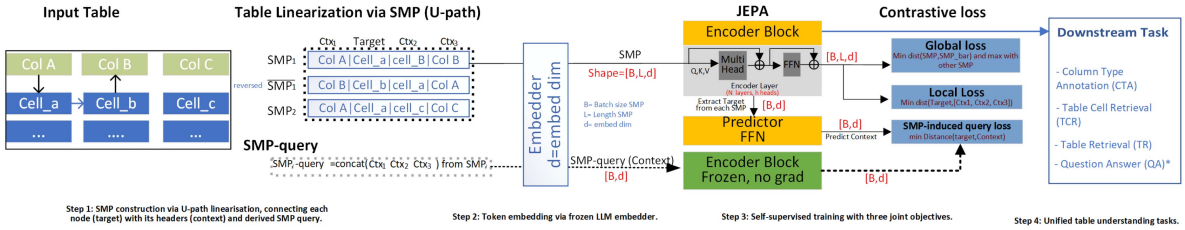


Fig. 1. The UTUEL framework introduces *TRL-JEPA*, a Joint-Embedding Predictive Architecture for table representation learning (TRL), in four steps. **(Step 1)** Each cell is linearized by U-path traversal into a Semantic Meta-Path (SMP) linking it to its row and column headers, with a context-only SMP-query context $=$ concat$(\mathrm{Ctx}_1, \mathrm{Ctx}_2, \mathrm{Ctx}_3)$ constructed in parallel. **(Step 2)** A frozen LLM embeds each SMP token into a $[B, L, d]$ representation and the SMP-query into a $[B, d]$ vector (B batch size, L SMP length, d embedding dimension). **(Step 3)** Three objectives are optimized jointly: an *encoder* refines the SMP embeddings under *global repulsion* (SMP against $\overline{\mathrm{SMP}}$) and *local alignment* (target cell with its context tokens), while a *predictor* FFN maps the target hidden state into a *frozen* encoder's space via the SMP-query (the SMP-induced query loss). Intuitively, the encoder learns a cell representation $h_c = \mathrm{Enc}(\mathrm{SMP}(c))$ whose context it can predict, maximizing $p(h_{\mathrm{context}} \mid h_c)$ in embedding space rather than generating text. **(Step 4)** The learned embeddings transfer, without a generative language head, to CTA, TCR, TR, and label-free QA, all answered by nearest-neighbour retrieval.

recent open-source general-purpose LLMs and table-specialized fine-tuned models.

II. Contributions

We make the following contributions:

1) *TRL-JEPA, a unified table embedding framework:* We formulate Table Representation Learning (TRL) as a Joint-Embedding Predictive Architecture (JEPA) [21], pairing an encoder with a predictor, trained over U-shaped SMPs that link each cell to its row and column context.
2) *Hierarchical TRL axioms for TU:* We introduce a context-aware learning objective that addresses the three core TU challenges: SMP-based contextualization with *local alignment* defines clear cell semantics, mitigating semantic complexity; *global repulsion* separates context-dependent meanings of the same surface value, capturing *contextual variability*; and layout-aware SMP linearization accommodates *structural heterogeneity*.
3) *Geometry-induced table cell retrieval:* Through an SMP-induced objective, the predictor maps inference-time queries into the learned embedding space, enabling online TQA by nearest-neighbour search without prompts or QA fine-tuning.
4) *A real-world table benchmark:* We release a domain-expert-curated TQA dataset of adhesive-datasheet tables, supporting cell-level retrieval within broader TU evaluation.

III. Background

TU models have followed the same architectural shift as NLP, moving from encoder-based designs such as BERT [22] and RoBERTa [23] towards decoder-based LLMs such as GPT [24] and DeepSeek [25]. Both families build on the Transformer [26], yet as the Observatory framework [9] highlights, selecting a model for a given TU task remains largely a matter of empirical trial and error rather than principled design, a gap our UTUEL aims to close. This section reviews the transition across three designs: (1) encoder-only, (2) decoder-only, and (3) hybrid encoder-decoder models. We highlight some of the most influential approaches in each category and compare them with our method where relevant.

A. Encoder-Only Models

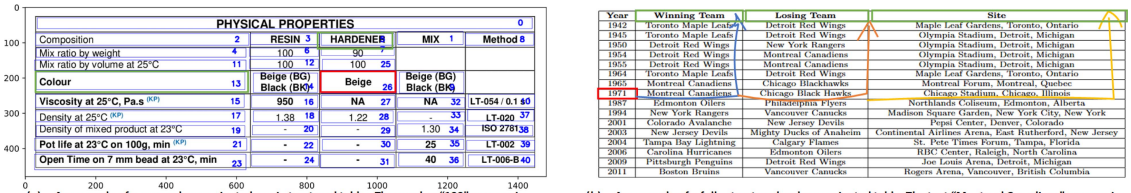
Encoder-only architectures such as TAPAS [27], TURL [28], TaBERT [29], HyTREL [30], Ditto [31], and Doduo [32] perform strongly on CTA, row-column annotation, and TQA, but transfer poorly across tasks. HyTREL and TURL, for instance, target CPA and Table Similarity Search (TS) and do not support TQA, whereas TAPAS, built for QA, does not extend to CPA or TS. For TQA, such models extract answer spans from row-linearized text, which suits look-up queries but cannot be transferred to other tasks such as CTA.

B. Decoder-Only Models

Decoder-only architectures have become the dominant generative approach for tables [33], [34], [35], [36], [37]. One line of work fine-tunes generative models on table data, as in Table-GPT2 [38], GritLM [20]; another relies on prompt engineering, as in DB-GPT [39] and [40], with Trompt [41] and TabLLM [42] further refining this direction, and Chain-of-Table [43] introducing multi-step reasoning. These methods treat tables as flat text obscuring structural and semantic relationships, a loss the LLM's capacity largely compensates for through its native NLP strength. Generalisation to unseen samples may thus stem mainly from the LLM rather than from any structural modelling of the table. The trade-off is cost: each query is re-encoded from scratch, fine-tuning or prompt-tuning demands substantial compute, and outputs can still hallucinate, requiring validation that limits full automation.

C. Encoder-Decoder Models

A third, hybrid line treats tables as images, combining a visual encoder (e.g., ResNet [44] or Vision Transformer [45]) with a text decoder, as in TableVLM [4] and TabPedia [46]. Such end-to-end image-to-answer models require extensive, high-quality supervision, confining evaluation to small curated subsets of datasets such as WTQ [47], FinTabNET [48], and TablePub1M [49], and generalizing poorly beyond them. We propose a lighter, modular alternative: instead of learning the table end-to-end from pixels, a dedicated table structure recognition model first recovers the cell grid, which UTUEL then consumes as structured input. We follow this modular pipeline



PHYSICAL PROPERTIES				
Composition	RESIN	HARDENER	MIX	Method
Mix ratio by weight	100	90		
Mix ratio by volume at 25°C	100	100		
Colour	Beige (BG) Black (BK)	Beige	Beige (BG) Black (BK)	
Viscosity at 25°C, Pa.s (KP)	950	NA	NA	LT-054 / 0.1 s0
Density at 25°C (KP)	1.38	1.22	-	LT-020
Density of mixed product at 23°C	-	-	1.30	ISO 2781
Pot life at 23°C on 100g, min (KP)	-	-	25	LT-002
Open Time on 7 mm bead at 23°C, min	-	-	40	LT-006-B

(a) An example of a row-column-oriented semi-structured table. The number "100" appears in multiple locations, but its in-context meaning varies depending on the table primitives, such as headers. It can be interpreted as the weight ratio of the RESIN or the volume ratio of the HARDENER.

Year	Winning Team	Losing Team	Site
1942	Toronto Maple Leafs	Detroit Red Wings	Maple Leaf Gardens, Toronto, Ontario
1945	Toronto Maple Leafs	Detroit Red Wings	Olympia Stadium, Detroit, Michigan
1950	Detroit Red Wings	New York Rangers	Olympia Stadium, Detroit, Michigan
1954	Detroit Red Wings	Montreal Canadiens	Olympia Stadium, Detroit, Michigan
1955	Detroit Red Wings	Montreal Canadiens	Olympia Stadium, Detroit, Michigan
1964	Toronto Maple Leafs	Detroit Red Wings	Maple Leaf Gardens, Toronto, Ontario
1965	Montreal Canadiens	Chicago Blackhawks	Montreal Forum, Montreal, Quebec
1971	Montreal Canadiens	Chicago Black Hawks	Chicago Stadium, Chicago, Illinois
1987	Edmonton Oilers	Philadelphia Flyers	Northlands Coliseum, Edmonton, Alberta
1994	New York Rangers	Vancouver Canucks	Madison Square Garden, New York City, New York
2001	Colorado Avalanche	New Jersey Devils	Pepsi Center, Denver, Colorado
2003	New Jersey Devils	Mighty Ducks of Anaheim	Continental Airlines Arena, East Rutherford, New Jersey
2004	Tampa Bay Lightning	Calgary Flames	St. Pete Times Forum, Tampa, Florida
2006	Carolina Hurricanes	Edmonton Oilers	RBC Center, Raleigh, North Carolina
2009	Pittsburgh Penguins	Detroit Red Wings	Joe Louis Arena, Detroit, Michigan
2011	Boston Bruins	Vancouver Canucks	Rogers Arena, Vancouver, British Columbia

(b) An example of a fully-structured, column-oriented table. The text "Montreal Canadiens" appears in multiple locations, but its in-context meaning depending on the table primitives, such as headers, are conflicting. It can be interpreted as the winning team in some years and the losing team in others.

Fig. 2. How table structure shapes a cell's canonical meaning. **(a)** The anchor entry "Beige" (red border) has a single abstract meaning tuple (Color, Hardener, Physical Property), captured by a Semantic Meta-Path (SMP) that traces an $\lrcorner$-shaped path through its adjacent headers. **(b)** The anchor entry "1971" (red border) is polysemous, carrying conflicting tuples such as (Year, Winning Team, Montreal Canadiens) versus (Year, Losing Team, Chicago Blackhawks), each from a distinct header relationship and SMP. UTUEL fine-tunes the LLM cell embedding into multiple context-specific instances, one per meaning. Each SMP follows a dual-path traversal that links the anchor cell to its row and column headers, enabling header-aware contrastive learning to disentangle polysemous cells.

to build our adhesive-datasheet benchmark, whose tables are cropped from complex document layouts and processed through table structure recognition as structured input.

In summary, these three families share similar limitations. Encoder-only models are typically tied to a specific task; decoder-only models recover table structure only implicitly, at a recurring per-query cost; and image-based encoder-decoder models depend on heavily curated supervision. Moreover, these approaches often rely on token embeddings learned over flattened table text, which lacks the grammatical order of natural language and can therefore misrepresent cell meaning. UTUEL instead preserves the semantic information already present in pretrained embeddings while adding table-specific meaning on top, by projecting header and entry cells into a refined space that captures cell-header relationships. The result is a single, label-free cell embedding computed once per table and reused at no extra cost across further queries and downstream tasks, including TCR, TQA, CTA, and TR. This offers an efficient, scalable alternative to the task-specific, per-query, and supervision-heavy TU paradigms above.

D. Table Cell Retrieval via Semantic Parsing

We survey the table cell retrieval methods that are supported by our cell-level embedding. Early work of [47] introduced KG-style semantic parsing that converts queries into logical forms to rank and retrieve cells. Subsequent methods incorporated deep learning through cosine-similarity cell scoring [50] and an iterative search optimising marginal likelihood over logical forms [51]. The Row-Column-Intersection (RCI) framework [52] and its refinement CLTR [53] instead retrieve cells through dual classification; table-question pairs are encoded as query-row and query-column sequences, scored by separate binary classifiers, and combined to identify candidate cells. In the absence of a dedicated cell-retrieval benchmark, RCI is evaluated on repurposed TQA datasets such as WikiTableQuestions, WikiSQL, and MCTQA. RCI also inspired TACR [54] for HybridQA [55], which combines RCI's top-k candidate cells with a discriminative model for joint table-text reasoning.

We note that these methods retrieve cells by ranking row- and column-level classifications, rather than by learning an explicit cell-level semantic embedding that can be transferred across TU tasks. UTUEL instead learns cell embeddings directly from each table, without query-answer annotations, and computes them once per table. At inference time, queries are answered online through nearest-neighbor search in the learned embedding space, without per-query fine-tuning. To support this paradigm, we introduce a real-world TQA dataset of adhesive-datasheet tables, built with domain experts to promote cell-level retrieval within broader TU.

IV. Methodology

This section presents UTUEL, our framework for learning context-aware cell embeddings. Fig. 3 serves throughout as a qualitative illustration of the TRL axioms.

A. The TRL Axioms

Let T be the set of elements (cells, row headers, column headers) of an $n \times m$ table:

$$T = \{c_{ij}\} \cup \{rh_i\} \cup \{ch_j\}, \quad i \in [n],\ j \in [m].$$

We formulate TRL as learning a refined space D_z from an initial space D_y:

$$T \xrightarrow{\Phi} D_y \xrightarrow[e_\phi, e_\theta]{\text{TRL}} D_z, \tag{1}$$

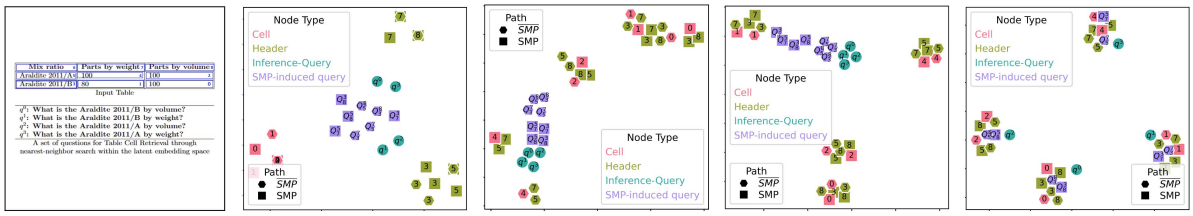
where $D_y = \{\Phi(x) \mid x \in T\}$, Φ is an LLM embedding function, and e_θ, e_ϕ are learnable encoders for cells and headers. The axioms then become an optimization problem.

Local alignment minimizes the distance $D(\cdot,\cdot)$ between each cell c_{ij} and its context. In Fig. 2(a), "Beige" should align with its tuple (HARDENER, COLOR) in D_z:

$$\min_\theta \left\{ \sum_{i,j} D\left(e_\theta(c_{ij}),\, e_\phi(rh_i)\right) + D\left(e_\theta(c_{ij}),\, e_\phi(ch_j)\right) \right\}. \tag{2}$$

Global repulsion pushes apart cells that, despite similar text, carry different meanings; in Fig. 2(b), "Montreal Canadiens" in 1955 (a loss) differs from 1971 (a win). It maximises pairwise distances between rows and between headers:

$$\max_\phi \left\{ \sum_{i' \neq i} D\left(e_\phi(rh_i),\, e_\phi(rh_{i'})\right) + \sum_{j' \neq j} D\left(e_\phi(ch_j),\, e_\phi(ch_{j'})\right) \right\}. \tag{3}$$



(a) Row-column-oriented table **(b)** Initial embedding **(c)** With local alignment **(d)** With global repulsion **(e)** With SMP-induced loss

Fig. 3. Embedding evolution under the hierarchical contrastive objectives of UTUEL, visualized as 2D t-SNE projections of cell and query embeddings. Colours indicate node type; marker shapes indicate path direction, normal (SMP) versus reverse ($\overline{\text{SMP}}$). (a) Input table, with cell indices in blue. We note that Cell 6 (the mix ratio) is a header cell, tied to the column headers ("by volume", "by weight") rather than its row header, an irregular layout that reflects the structural heterogeneity we aim to disentangle. UTUEL handles such an ambiguous header naturally by design: Cell 6 is never a target of an $\mathfrak{Z}$-shaped SMP, so it does not appear in the embedding in (b–e). (b) Initial LLM embeddings separate node types poorly, overlapping distinct cells (e.g. 0, 2, 4). (c) The cell-level loss (local alignment) clusters each cell with its header context, with some inter-type overlap. (d) The SMP-level loss (global repulsion) sharpens separation between node types. (e) The query-level loss aligns SMP-induced queries with their answer cells, enabling nearest-neighbour search.

Jointly optimising (2) and (3) is ideal but ill-posed: e_ϕ and e_θ receive conflicting gradients, as repulsion pushes headers apart while alignment pulls them towards cells, a representational "tug-of-war".

B. Semantic Meta-Paths (SMPs)

We resolve this tug-of-war by interpreting the TRL principles *hierarchically*. Using the Semantic Meta-Path (SMP) framework [11], each cell is bound to its row and column headers within a single context path. This lets *cell-level alignment* (within an SMP) and *SMP-level repulsion* (across SMPs) be optimised jointly by the TRL-JEPA encoder Ψ_θ, with the predictor p_ψ introduced later for query retrieval. The SMP S_{ij} anchored at c_{ij} with context $X_{S_{ij}} = \{rh_i, ch_j\}$ is

$$S_{ij} = \{rh_i,\, c_{i,1},\, \ldots,\, c_{i,j},\, c_{i-1,j},\, \ldots,\, c_{1,j},\, ch_j\},$$

with D_y representation

$$S^y_{ij} = \{y^{rh_i}_{S_{ij}}, \ldots, y^{c_{ij}}_{S_{ij}}, \ldots, y^{ch_j}_{S_{ij}}\}, \quad X^y_{S_{ij}} = \{y^{rh_i}_{S_{ij}}, y^{ch_j}_{S_{ij}}\},$$

where $y^{(\cdot)}_{S_{ij}} = \Phi(\cdot)$. Each S_{ij} has a reverse variant $\bar{S}_{ij}$ with the same anchor but reversed order ($\bar{\bar{S}}_{ij} = S_{ij}$), giving

$$G = \{S_{ij},\, \bar{S}_{ij} \mid i \in [n],\, j \in [m]\}. \tag{4}$$

A single transformer-based *TRL encoder* Ψ_θ operates over SMP sequences, subsuming e_θ and e_ϕ and yielding refined embeddings in D_z:

$$S^z_{ij} = \Psi_\theta(S^y_{ij}) = \{Z^{rh_i}_{S_{ij}}, \ldots, Z^{c_{ij}}_{S_{ij}}, \ldots, Z^{ch_j}_{S_{ij}}\},$$
$$X^z_{S_{ij}} = \{Z^{rh_i}_{S_{ij}}, Z^{ch_j}_{S_{ij}}\}.$$

At the SMP level, *local alignment* draws the anchor cell towards its context X while separating it from unrelated cells in the same SMP:

$$\min_\theta \sum_{S_{ij} \in G} \left\{ \sum_{Z_{S_{ij}} \in X^z_{S_{ij}}} D\left(Z^{c_{ij}}_{S_{ij}},\, Z_{S_{ij}}\right) + \frac{1}{\sum\limits_{\substack{Z_{S_{ij}} \in S^z_{ij} \\ Z_{S_{ij}} \notin X^z_{S_{ij}} \cup \{Z^{c_{ij}}_{S_{ij}}\}}} D\left(Z^{c_{ij}}_{S_{ij}},\, Z_{S_{ij}}\right)} \right\}. \tag{5}$$

Global repulsion keeps distinct SMPs apart while holding each close to its reverse variant:

$$\min_\theta \sum_{S_{ij} \in G} \left\{ D\left(Z^{rh_i}_{S_{ij}},\, Z^{rh_i}_{\bar{S}_{ij}}\right) + D\left(Z^{ch_j}_{S_{ij}},\, Z^{ch_j}_{\bar{S}_{ij}}\right) + \frac{1}{\sum\limits_{\substack{S_{kl} \in G \\ S_{kl} \neq S_{ij},\, \bar{S}_{ij}}} \left[D\left(Z^{rh_i}_{S_{ij}},\, Z^{rh_k}_{\bar{S}_{kl}}\right) + D\left(Z^{ch_j}_{S_{ij}},\, Z^{ch_l}_{\bar{S}_{kl}}\right) \right]} \right\}. \tag{6}$$

The joint SMP-level objective combines (5) and (6).

C. Learning Objectives

Both objectives fit the InfoNCE framework [56]:

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E}\left[\log \frac{\langle z, z^+ \rangle}{\langle z, z^+ \rangle + \sum_{z^- \in Z^-} \langle z, z^- \rangle}\right], \tag{7}$$

where $\langle \cdot, \cdot \rangle = \exp(\text{sim}(\cdot, \cdot)/\tau)$, z is the anchor, z^+ its positive, Z^- negatives set, sim is cosine similarity, and τ a temperature. Applying $\log(a/b) = \log a - \log b$, our objectives instantiate this form.

1) Local Alignment Objective:

$$\mathcal{L}_{\text{Local}}(\theta) = -\sum_{S_{ij} \in G} \left[\log \sum_{Z_{S_{ij}} \in X^z_{S_{ij}}} \langle Z^{c_{ij}}_{S_{ij}},\, Z_{S_{ij}} \rangle \right.$$

$$- \log \left(\sum_{Z_{S_{ij}} \in X^z_{S_{ij}}} \langle Z^{c_{ij}}_{S_{ij}}, Z_{S_{ij}} \rangle + \sum_{\substack{Z_{S_{ij}} \in S^z_{ij} \\ Z_{S_{ij}} \notin X^z_{S_{ij}} \cup \{Z^{c_{ij}}_{S_{ij}}\}}} \langle Z^{c_{ij}}_{S_{ij}}, Z_{S_{ij}} \rangle \right) \Bigg] . \quad (8)$$

2) Global Repulsion Objective:

$$\mathcal{L}_{\text{rep}}(\theta) = - \sum_{S_{ij} \in G} \Bigg[\log \left(\left\langle Z^{rh_i}_{S_{ij}}, Z^{rh_i}_{\bar{S}_{ij}} \right\rangle + \left\langle Z^{ch_j}_{S_{ij}}, Z^{ch_j}_{\bar{S}_{ij}} \right\rangle \right) - \log \left(\left\langle Z^{rh_i}_{S_{ij}}, Z^{rh_i}_{\bar{S}_{ij}} \right\rangle + \left\langle Z^{ch_j}_{S_{ij}}, Z^{ch_j}_{\bar{S}_{ij}} \right\rangle + \sum_{\substack{S_{kl} \in G \\ S_{kl} \neq S_{ij}, \bar{S}_{ij}}} \langle Z^{rh_i}_{S_{ij}}, Z^{rh_k}_{\bar{S}_{kl}} \rangle + \langle Z^{ch_j}_{S_{ij}}, Z^{ch_l}_{\bar{S}_{kl}} \rangle \right) \Bigg] . \quad (9)$$

3) SMP-Induced Learning: The local alignment and global repulsion objectives train the encoder Ψ_θ, which fixes the embedding space D_z in which cells are organised. Following our TRL-JEPA architecture, we now use the TRL-JEPA *predictor* p_ψ, which maps a query into this same space, reducing inference to locating the query near its target cell. A query is matched to a cell through its context: consider *"What is the color of the hardener?"* in Fig. 2(a), whose *High-Information Tokens (HITs)* "color" and "hardener" correspond to the contexts $X_{S_{ij}} = \{rh_i, ch_j\}$ of the SMP anchored at the target cell "Beige". The predictor combines these headers into a single composite representation and aligns the cell with it via a sigmoid contrastive loss [57]:

$$\mathcal{L}_{\text{HIT}}(\psi, \theta) = - \sum_{S_{ij} \in G} \log \sigma \left(\text{sim}(Y_{\text{HIT}}, p_\psi(Z^{c_{ij}}_{S_{ij}})) \right) , \quad (10)$$

where $\sigma(x) = 1/(1 + \exp(-x))$ and

$$Y_{\text{HIT}} = \Psi_{\bar{\theta}} \left(\Phi \left(\bigcup X_{S_{ij}} \right) \right) = \Psi_{\bar{\theta}}(\Phi(rh_i + ch_j))$$

is the LLM embedding of the SMP-induced query HIT, with $\Psi_{\bar{\theta}}$ being the frozen TRL-JEPA encoder. Combining (8), (9), and (10) gives the unified TRL objective over the encoder and predictor:

$$\min_{\theta, \psi} \mathcal{L} = \frac{1}{|G|} \left(\alpha \, \mathcal{L}_{\text{local}}(\theta) + \beta \, \mathcal{L}_{\text{rep}}(\theta) + \gamma \, \mathcal{L}_{\text{HIT}}(\psi, \theta) \right) , \quad (11)$$

where $\alpha, \beta, \gamma \in [0, 1]$ weight each loss. The objective induces a coherent, sparse geometry in which cosine similarity reflects semantic relationships, so nearest-neighbour search (dense passage retrieval [58]) retrieves the cell for a given query q:

$$\text{answer}(q) = \arg \max_{c_{ij} \in T} \text{sim} \left(\Psi_{\bar{\theta}}(\Phi(q)), Z^{c_{ij}}_{S_{ij}} \right) . \quad (12)$$

Qualitative illustration: Fig. 3 illustrates how the loss components jointly address the TU challenges of Section I: *local alignment* (8) addresses semantic complexity, *global repulsion* (9) addresses contextual variability, and the SMP linearisation disentangles structural heterogeneity. The SMP-induced predictor term (10) completes the unified objective (11), yielding a geometry where nearest-neighbour search (12) retrieves the target cell for a query. In the experiments, we report performance under different query-filtering modes to assess robustness to query distribution shift.

V. Experiments

We evaluate UTUEL against strong LLM baselines across four TU tasks. On our newly introduced AdhesiveDataset, we assess TCR and online TQA. On a popular public benchmark, we evaluate TQA and, by reusing the learned embeddings without retraining for TR. We further evaluate CTA on TURL-CTA [28]. In addition, we test robustness under query-distribution shifts, comparing raw, stop-word-filtered, and HIT-based queries, and ablate the TRL-JEPA loss terms.

We report three metrics: *Exact Match* (EM) accuracy for prompted LLMs, and Hit@(k) and Mean Reciprocal Rank (MRR) for retrieval-based methods. Hit@(k) measures whether the correct answer appears among the top-(k) retrieved cells, while MRR is the average inverse rank of the correct answer, assigned 0 when the answer is not retrieved. The full evaluation metrics, task formulation, baseline model, prompt baseline templates, and training hyper-parameters, are further elaborated in Appendix Sections 3, 4, 5, 6, and 7, respectively.

A. Efficient SMP Generation

For a table of n rows and m columns, generating SMPs for every cell costs $\mathcal{O}(nm)$: each target cell is linearized into two U-shaped paths of the form $header_a \rightarrow node_a$ (target) $\rightarrow node_b \rightarrow header_b$, one traversing its row and one its column (e.g., the center cell of a 3×3 table yields two such paths), and this cost grows with table size.

TU models often mitigate this cost by truncating rows beyond a fixed threshold. While acceptable for CTA, where a column's type may be inferable from a few rows, this truncation is unsuitable for TQA, where the answer may lie in a discarded row. We therefore adopt a divide-and-conquer SMP-generation strategy that preserves all rows. Specifically, a large table is partitioned into overlapping chunks of k consecutive rows ($k \ll n$); each chunk is treated as a subgraph that repeats the header row to retain column semantics. SMPs are then computed locally within each chunk, and the resulting paths are merged into the global SMP set. Because each chunk spans only k rows, the maximum path length is bounded locally, reducing the per-chunk generation cost to $\mathcal{O}(km)$ while still covering all rows across the $\lceil n/k \rceil$ chunks. The full SMP generation algorithm and complexity analysis are provided in Section 2 in the Appendix.

TABLE I
PERFORMANCE (% ± STD) OF VARIOUS LLMS ON ADHESIVETABLEQA ACROSS SERIALIZATION FORMATS (SMP, JSON, FLATTEN). METRICS (AVERAGED OVER FIVE RUNS): ACCURACY (EXACT MATCH) OR HIT@1 FOR UTUEL; PARSING ERROR (PERCENTAGE OF RESPONSES THAT FAILED TO PARSE INTO THE REQUIRED FORMAT); AND #TOKENS (TOTAL INPUT PLUS OUTPUT TOKENS OVER ALL 816 QUESTIONS).

Method/LLM (#param)	Serialisation	Accuracy Hit@1	Parsing Error	#Tokens
TableGPT2** (7B)	JSON	43.68 ± 0.70	3.62 ± 0.43	0.771M
	SMP	41.10 ± 1.15	5.04 ± 0.24	0.625M
	Flatten	70.09 ± 0.75	2.72 ± 0.50	0.112M
GritLM** (7B)	JSON	16.61 ± 0.45	53.30 ± 1.23	0.726M
	SMP	15.65 ± 0.00	61.45 ± 0.00	0.625M
	Flatten	51.13 ± 0.06	26.90 ± 0.52	0.107M
Llama3 (8B)	JSON	20.23 ± 0.98	22.32 ± 0.70	0.965M
	SMP	26.34 ± 1.74	14.45 ± 1.10	1.032M
	Flatten	51.97 ± 0.63	12.93 ± 0.74	0.158M
DeepSeek-r1 (8B)	JSON	46.23 ± 0.72	24.64 ± 0.38	0.763M
	SMP	34.35 ± 1.85	28.84 ± 1.94	0.624M
	Flatten	61.35 ± 0.80	14.61 ± 0.17	0.112M
Gemma2 (9B)	JSON	47.49 ± 0.74	4.61 ± 0.56	0.788M
	SMP	40.72 ± 0.48	13.67 ± 0.24	0.632M
	Flatten	68.45 ± 0.27	6.67 ± 0.34	0.129M
GPT-OSS (20B)	JSON	68.93 ± 1.04	1.48 ± 0.28	0.762M
	SMP	41.42 ± 0.61	3.48 ± 0.66	0.625M
	Flatten	64.41 ± 1.00	5.91 ± 0.38	0.113M
Qwen3.6 (27B)	JSON	73.44 ± 0.69	6.12 ± 0.90	0.763M
	SMP	62.95 ± 1.69	11.88 ± 1.85	0.623M
	Flatten	73.82 ± 0.44	1.84 ± 0.08	0.138M
Gemma4 (27B)	JSON	69.98 ± 3.45	8.62 ± 4.77	0.763M
	SMP	47.68 ± 5.95	28.81 ± 9.10	0.623M
	Flatten	73.09 ± 1.04	2.63 ± 0.99	0.150M
	Init. Embedder			
UTUEL (ours)	all-MiniLM (22M)	75.55	–	**0.011M**
1-layer, 12-head	all-MiniLM-L6-v2† (22.7M)	75.55	–	**0.011M**
model size	all-mpnet-base-v2† (110M)	74.69	–	**0.011M**
2.9M @ dim 384	Nomic-embed-text (270M)	36.24	–	**0.011M**
11.8M @ dim 768	EmbeddingGemma (300M)	**81.57**	–	**0.011M**
	Qwen3-Embedding (0.6B)	74.32	–	**0.011M**

† HuggingFace sentence-transformer; other embedders are LLMs served via Ollama; ** fine-tuned on tables.

TABLE II
PERFORMANCE (% ± STD) ON THE SQL LOOKUP TEST SET (11,324 TABLES), AVERAGED PER TABLE OVER FIVE ITERATIONS. CATEGORY 1 PROMPTS LLMS ONLINE (NO FINE-TUNING); CATEGORY 2 IS PRETRAINED OR FINE-TUNED ON THE DATASET. ACCURACY IS EXACT MATCH OR HIT@1, PARSING ERROR (PERCENTAGE OF RESPONSES THAT FAILED TO PARSE INTO THE REQUIRED FORMAT), AND #TOKENS IS THE TOTAL INPUT AND OUTPUT TOKENS OVER THE DATASET.

Models (Size)	MRR	Accuracy Hit@1	Parsing Error	#tokens
Category 1: LLM-based (LLM prompting)				
TableGPT2** (7B)	–	78.22 ± 0.19	1.05 ± 0.05	3.59M
GritLM** (7B)	–	40.13 ± 1.03	41.09 ± 1.41	3.59M
TableLLM** (7B-Q4)	–	22.9	61.5	3.27M
LLaMA3 (8B)	–	54.83 ± 0.41	8.16 ± 0.25	3.85M
DeepSeek-r1 (8B)	–	70.81 ± 0.14	7.88 ± 0.18	3.58M
Gemma2 (9B)	–	68.45 ± 0.12	2.09 ± 0.06	3.82M
GPT-OSS (20B)	–	77.58 ± 0.07	2.71 ± 0.09	3.58M
Qwen3.6 (27B)	–	**87.22**	1.65	4.00M
Gemma4 (27B)	–	85.31 ± 1.47	3.17 ± 1.84	4.33M
Ours* (2.9M – 11.8M)				
EmbeddingGemma (300M)	**0.767**	**66.73**	–	**0.34M**
Qwen3-Embedding (0.6B)	0.709	59.71	–	**0.34M**
all-MiniLM (22M)	0.674	56.60	–	**0.34M**
Category 2: Pretrained or fine-tuned on the dataset				
$RCI^{\dagger}$ (75M)	**0.962**	**94.60**	–	–
TaBERT (131M)	0.761	71.16	–	–
TAPAS (110M)	–	89.02	–	–
TAPAS (340M)	–	89.49	–	–

–: not applicable; ** fine-tuned on tables; * UTUEL encoder: 384/768 dims, 12 heads, 1 layer; †: fully supervised.

TABLE III
ABLATION ON WIKISQL-LOOKUP (MRR) OVER ALL COMBINATIONS OF THE THREE LOSS TERMS, LOCAL ALIGNMENT $\mathcal{L}_{\text{LOCAL}}$, GLOBAL REPULSION $\mathcal{L}_{\text{GLOBAL}}$, AND THE HIT OBJECTIVE $\mathcal{L}_{\text{HIT}}$, ACROSS FIVE INITIAL EMBEDDERS

Loss activated			MRR by embedder				
$\mathcal{L}_{\text{Local}}$ α	$\mathcal{L}_{\text{Global}}$ β	$\mathcal{L}_{\text{HIT}}$ γ	all-MiniLM	Emb. Gemma	nomic-emb.	Qwen3-Emb.	MiniLM-L6-v2
		✓	0.063	0.081	0.065	0.073	0.068
	✓	✓	0.049	0.048	0.055	0.042	0.056
✓			0.662	0.748	0.199	0.698	0.086
✓		✓	0.630	0.748	0.187	0.672	0.104
✓	✓		0.654	0.753	0.193	**0.709**	0.085
✓	✓	✓	**0.674**	**0.767**	**0.220**	0.647	**0.131**

B. Table Cell Retrieval on Our New Table Dataset

We introduce *AdhesiveTableQA*, a real-world TCR benchmark consisting of 816 expert-validated questions for adhesive-material selection. Developed with domain experts, the benchmark targets the time-consuming task of precisely locating table cells [59] and supports practical use cases such as property matching, multi-criteria comparison, and industrial material selection. Full details on dataset construction, annotation protocol, and statistics are provided in Section 1 of the Appendix.

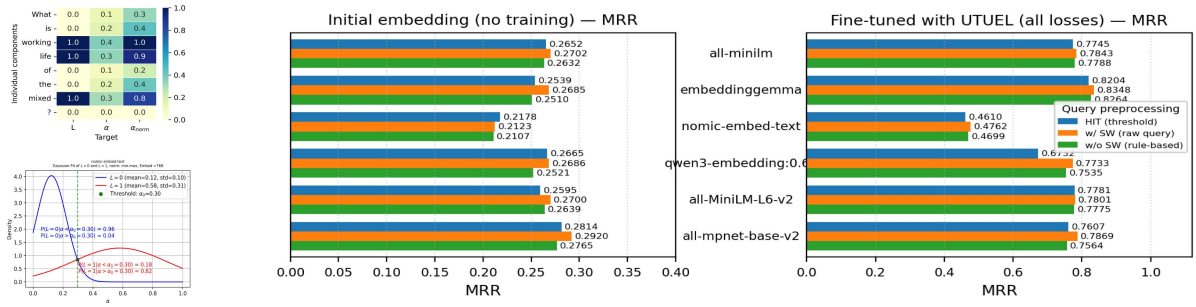
In Table I, we report the performance of UTUEL against table-specialized fine-tuned models and general-purpose LLMs prompted on linearized tables, each serialized in three formats: JSON, Semantic Meta-Path (SMP) [11], and a flattened form.

1) Performance Analysis:

a) LLM baselines: Serialisation format strongly affects accuracy: flattened text-like inputs score highest for most models, and among structured formats JSON generally outperforms SMP, with accuracy increasing with model size. Two artefacts complete the picture beyond accuracy. The *parsing error* measures how often a response deviates from the requested format, reported per model and serialisation in Table I. Exact Match also penalises answers that are correct but differ by special characters, units, or formatting; to separate these surface mismatches from genuine errors, we report the cosine similarity between gold and extracted answers in Fig. 5 in Section 8 of the Appendix.

b) UTUEL: Unlike prompted LLM baselines, which re-encode the full table for every query, UTUEL computes a context-aware embedding for each table *once* (11) and answers any subsequent query by nearest-neighbour search in the cached space (12), without prompting or task-specific fine-tuning. With a lightweight encoder of only $2.9M - 11.8M$ parameters over frozen embeddings, UTUEL reaches 81.57% accuracy with EmbeddingGemma, remaining competitive with the strongest prompted LLMs while avoiding their repeated per-query token cost. The main exception is Nomic-embed-text, which achieves only 36.24%; its weaker initial embedding geometry appears difficult for the lightweight encoder to restructure, consistent with its behavior across our other experiments.

2) Query-Level Distribution Shift: We assess robustness to query wording across three modes, **RAW** (full query), **SW** (stop words removed), and **HIT-filtered** (only retaining tokens that most shape the query semantics). Per-embedder results are shown in Fig. 4(b). HITs are tokens whose score $\alpha_i = \text{sim}(\Phi(q), \Phi(q_i))$ exceeds a threshold α_0 fitted on 500 annotated queries (Fig. 4(a)). Accuracy remains largely stable under filtering, but accuracy alone does not reveal how far the query representation moves; Fig. 5 therefore measures the corresponding embedding shift. That HIT preserves accuracy while discarding most tokens indicates that the retained header-aligned tokens carry the main query signal. A loss ablation on a separate, larger dataset (Table III) then identifies which objective term drives this robustness.



(a) HIT-filtered using *nomic-embed-text*.

(b) MRR by LLM Embedders from the initial to the refined embedding space.

Fig. 4. (a) HIT-token analysis for the query "What is the working life of the mixed?". Top: token embedding scores on Nomic-embed-text, where each token is scored by α, the cosine similarity between the LLM embedding of the full query and that of the token; a binary label L is assigned per token ($L = 1$ for a HIT, $L = 0$ otherwise). Bottom: the corresponding distribution of the min-max-normalized scores α_{norm}, with threshold α_0 fitted as a Gaussian over 500 manually annotated queries; the curve's position relative to α_0 captures the accuracy–error trade-off of the HIT detector. (b) MRR across LLM embeddings before (initial, no training) and after UTUEL fine-tuning, under three query preprocessing modes: **RAW** keeps the full query ("*what is working life of the mixed*"), **SW** removes stop words ("*working life mixed*"), and **HIT** keeps only the high-information tokens above α_0 ("*what working life the mixed*"). UTUEL improves MRR for every embedder except nomic-embed-text, and the three modes stay close, indicating robustness to query filtering, examined further in the distribution-shift analysis of Fig. 5.



(a) Nomic-embed-text

(b) Qwen3-embedding:0.6B

Fig. 5. Per-model content-based query distribution shift over the 816 Adhesive queries. The left panel histograms the cosine similarity of stop-word-filtered (SW) and HIT-filtered queries to the raw query (fitted Gaussian, mean μ); the right panel shows a shared 2D PCA of the raw, SW, and HIT embeddings. Similarity near 1 and overlapping clouds indicate robustness to wording. HIT induces the larger shift (lower μ), keeping only the High-Information Tokens, largely the target cell's headers, while SW stays closer to the raw query, e.g. RAW "What is the Roller (Bell) Peel Strength for Cure cycle 1?" becomes SW "Roller (Bell) Peel Strength Cure cycle 1" and HIT "Peel Strength Cure cycle". This shift tracks MRR score (Fig. 4(b)): models that shift least (e.g. Nomic) retain accuracy under HIT, while those shifting more, largest for Qwen3-Embedding, degrade.

C. Online TQA on the WikiSQL Public Dataset

The WikiSQL look-up dataset introduced by IBM in the Row-Column-Intersection (RCI) method [52] comprises 11,324 table-question pairs over 4,242 unique tables (about three questions per table). UTUEL is compared against prompted general-purpose LLMs and table-specialised fine-tuned models, with results reported in Table II.

1) Performance Analysis: Two regimes emerge from Table II. Prompted general-purpose LLMs achieve the highest accuracy (e.g. Qwen3.6 at 87.2% and Gemma4 at 85.3%), at 7–27B parameters and 4–4.3 M tokens over the test set, since each query re-encodes the full table. UTUEL takes the opposite trade-off: it embeds each table *once* and answers any subsequent query by nearest-neighbour search at no further token cost. With a 2.9–11.8M-parameter encoder over a lightweight initial embedder (22M–600 M), UTUEL reaches Hit@1 up to 66.7% (MRR 0.767, EmbeddingGemma), trailing the strongest prompted models on raw accuracy but at a fraction of their size and inference cost. The gap to the fully supervised RCI (94.6%) reflects its task-specific training, whereas UTUEL remains label-free and reusable across tasks.

2) Component Ablation Study: Table III isolates each term of (11). Three patterns emerge. First, $\mathcal{L}_{\text{Local}}$ carries most of the contribution: alone it already reaches strong MRR (e.g. 0.748 on EmbeddingGemma), and adding $\mathcal{L}_{\text{Global}}$ or $\mathcal{L}_{\text{HIT}}$ yields only marginal gains. Second, without $\mathcal{L}_{\text{Local}}$ the model drops to near-random (≤ 0.08): $\mathcal{L}_{\text{Local}}$ aligns each cell with its headers directly in the encoder space and establishes the geometry the other terms build on, whereas $\mathcal{L}_{\text{HIT}}$ acts only after the predictor projection and sharpens query–cell matching without organising the space, and $\mathcal{L}_{\text{Global}}$ likewise presupposes Local's alignment. Because a look-up query is composed of the target cell's headers, $\mathcal{L}_{\text{Local}}$ already encodes the query-to-target mapping, and $\mathcal{L}_{\text{HIT}}$ only refines it, a behaviour we examine by query category in Fig. 6. Third, the effect is embedder-dependent: gains are largest for well-structured initial spaces and smallest for nomic-embed-text, a dependence we analyse geometrically in Figs. 7 and 8.

3) Qualitative Analysis: The two geometric views explain the embedder-dependence in Table III. Fig. 7 confirms that $\mathcal{L}_{\text{Local}}$ creates the cluster structure. At the same time, the other terms refine it, and Fig. 8 shows the same effect per query: where the initial space is too weakly structured to reorganise

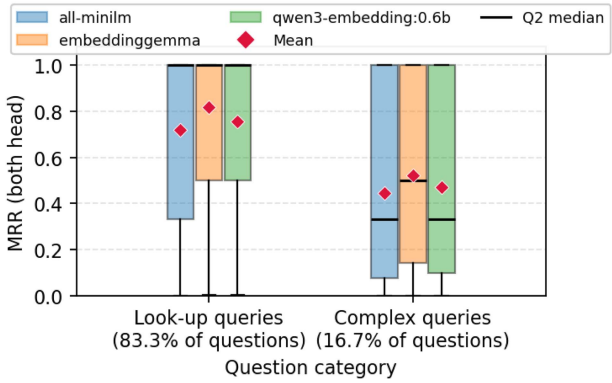


Fig. 6. Per-question MRR distribution on WikiSQL-Lookup by query category, across the five embedders. Look-up queries (83.3%) are single retrievals of the form "What is Joss'nationality?"; complex queries (16.7%) require comparison, aggregation, or arithmetic reasoning, e.g. "Who made the most scores in New Jersey?". In both categories, the answer is a single table cell, making the task suitable for Table Cell Retrieval.

TABLE IV
TABLE RETRIEVAL ON WIKISQL-LOOKUP (11,324 QUERIES OVER 4,242 TABLES). EACH TABLE EMBEDDING IS THE MEAN OF ITS CELL EMBEDDINGS; FOR EACH QUERY WE RETRIEVE THE TOP TABLES AND REPORT MRR AND HIT@k. UTUEL EMBEDDINGS (PRETRAINED) ARE COMPARED AGAINST THE SAME GENERAL-PURPOSE EMBEDDERS APPLIED DIRECTLY.

Model	MRR	Hit@1	Hit@3	Hit@5	Hit@10	Hit@20
UTUEL (2.9M – 11.8M) (ours, pretrained)						
EmbeddingGemma	**0.748**	**0.663**	**0.803**	**0.849**	**0.898**	**0.938**
Qwen3-Embedding	0.699	0.606	0.755	0.810	0.869	0.918
all-MiniLM	0.637	0.543	0.691	0.749	0.811	0.870
MiniLM-L6-v2	0.624	0.527	0.678	0.740	0.808	0.867
nomic-embed-text	0.123	0.079	0.132	0.161	0.208	0.265
General-purpose embedders (no UTUEL)						
EmbeddingGemma	0.423	0.328	0.463	0.526	0.608	0.686
Qwen3-Embedding	0.396	0.303	0.436	0.497	0.577	0.656
all-MiniLM	0.306	0.226	0.333	0.390	0.461	0.536
MiniLM-L6-v2	0.299	0.218	0.326	0.381	0.455	0.530
nomic-embed-text	0.127	0.084	0.136	0.164	0.206	0.256

(nomic-embed-text), answers remain unretrievable, consistent with its low MRR across the ablation.

D. Table Retrieval (TR)

To show that UTUEL transfers across tasks, we reuse the cell embeddings learned for TQA, without retraining, on Table Retrieval (TR), a core component of the Table-RAG framework. A table-level embedding is simply the mean of its already-computed cell embeddings, so for each query we retrieve the top tables by nearest-neighbour search at zero additional training cost (Table IV), comparing against the same general-purpose embedders applied directly. The effect is striking and consistent: fine-tuning any initial embedder with UTUEL roughly doubles retrieval MRR, EmbeddingGemma from 0.423 to 0.748, Qwen3-Embedding from 0.396 to 0.699, and all-MiniLM from 0.306 to 0.637, confirming that the learned cell-level geometry transfers without TR-specific supervision. The sole exception is nomic-embed-text ($0.127 \rightarrow 0.123$): consistent with its weak initial geometry in the loss ablation, UTUEL refines an existing semantic space but cannot manufacture one from a poor embedder. To the best of our knowledge, this is the first time TR is reported on this dataset in this reuse setting, and we release these results as a reference point for future work.

TABLE V
UTUEL CTA PERFORMANCE ON THE TURL-CTA DATASET (K = MAXIMUM ROWS USED PER COLUMN). WHILE FAR SMALLER THAN TABERT (131.9M) AND HYTREL (179.4M), UTUEL REMAINS COMPETITIVE AT MATCHED K, EXCEEDING TABERT$_{\text{BASE}}$ ON PRECISION AND TRAILING ONLY MARGINALLY ON F1. BEST UTUEL VALUES PER METRIC ARE IN **BOLD**.

Models	# param	F1/Precision/Recall (%)
TURL + metadata	–	92.79/93.25/92.34
Doduo + metadata	–	93.25/92.34/92.79
HYTREL w/o Pre-training ($K = 30$)	179.4M	92.71/92.50/92.92
HYTREL w/ Pre-training ($K = 30$)	179.4M	93.53/92.85/94.21
TaBERT$_{\text{base}}$ w/o Pre-train ($K = 3$)	131.9M	88.97/90.77/87.23
TaBERT$_{\text{base}}$ w/ Pre-train ($K = 3$)	131.9M	91.37/91.63/91.12
TaBERT$_{\text{base}}$ w/o Pre-train ($K = 1$)	131.9M	87.70/90.00/85.50
TaBERT$_{\text{base}}$ w/ Pre-train ($K = 1$)	131.9M	90.43/91.40/89.49
UTUEL (ours), w/ $\mathcal{L}_{\text{Local}}$, 12-head encoder		
all-minilm		
w/o pre-training ($K = 1$, 384-dim, 1-layer)	1.9M	88.47/87.55/89.41
w/ pre-training ($K = 1$, 384-dim, 3-layer)	5.4M	89.06/88.67/89.45
w/ pre-training ($K = 3$, 384-dim, 3-layer)	5.4M	90.20/91.42/89.01
all-MiniLM-L6-v2		
w/o pre-training ($K = 1$, 384-dim, 1-layer)	1.9M	88.52/87.59/89.47
w/ pre-training ($K = 1$, 384-dim, 3-layer)	5.4M	89.07/88.77/89.37
all-mpnet-base-v2		
w/o pre-training ($K = 1$, 768-dim, 1-layer)	7.2M	89.03/88.31/89.77
w/o pre-training ($K = 3$, 768-dim, 3-layer)	21.5M	**90.68/92.27**/89.14
w/ pre-training ($K = 1$, 384-dim, 1-layer)	1.9M	88.99/87.84/**90.17**
EmbeddingGemma		
w/o pre-training ($K = 1$, 768-dim, 1-layer)	7.2M	89.01/88.51/89.52
w/o pre-training ($K = 3$, 768-dim, 3-layer)	21.5M	90.57/92.09/89.11
Nomic-embed-text		
w/o pre-training ($K = 1$, 768-dim, 1-layer)	7.2M	86.18/86.18/86.17
w/ pre-training ($K = 1$, 768-dim, 1-layer)	7.2M	86.15/86.15/86.14
Qwen3-Embedding (0.6B)		
w/o pre-training ($K = 1$, 768-dim, 1-layer)	7.2M	89.05/88.38/89.73
w/o pre-training ($K = 3$, 768-dim, 3-layer)	21.5M	90.33/91.81/88.89

E. Column Type Annotation (CTA)

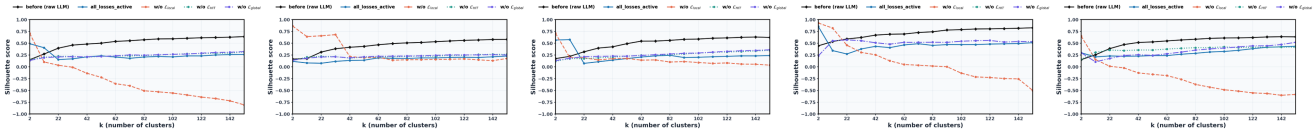
On the public TURL-CTA dataset, we classify each column with a lightweight fully connected head over UTUEL cell embeddings, and report performance with and without pretraining. Unlike prior work that uses the column header, UTUEL represents each column by its first non-header cell ($K = 1$), inferring the type from cell content rather than header text. Table V reports the results.

Fig. 9 shows that UTUEL groups cells of the same column type into colour-coherent clusters, absent in the initial LLM embedding. At a small fraction of the baselines' size (1.9–21.5 M versus TaBERT's 131.9 M and HYTREL's 179.4 M), UTUEL is competitive: at $K = 3$ it reaches 90.68 F1 and 92.27 Precision (all-mpnet-base-v2), exceeding TaBERT$_{\text{base}}$ on Precision and trailing only marginally on F1. Increasing K and pretraining both add consistent gains for the stronger embedders, making UTUEL well-suited to low-data, resource-constrained deployment.

F. Limitations and Future Work

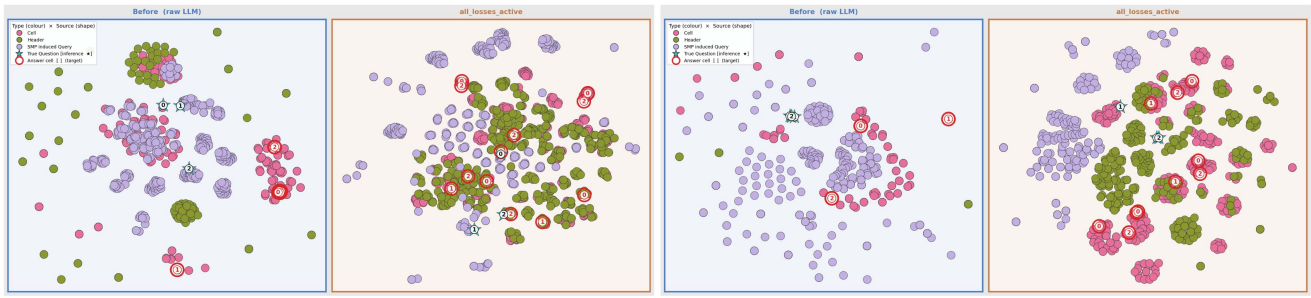
This work is one of the first proof-of-concept studies for learning general-purpose, context-aware cell-level table embeddings, akin to word embeddings in NLP, with several limitations.

a) Breadth of the analogy: We establish the cell-level analogue primarily on look-up TQA, TR, and CTA. An immediate next step is to demonstrate UTUEL's broader reusability across



(a) All-minilm (b) Embeddinggemma (c) Qwen3-embedding (d) Nomic-embed-text (e) all-MiniLM-L6-v2

Fig. 7. Silhouette score versus the number of clusters k for each loss combination, across the five embedders. From 10 randomly sampled tables (≥ 15 columns, ≥ 20 rows), 5,734 cell embeddings are clustered by k-means. Silhouette measures cluster *separability:* scores above 0.6 indicate compact, well-separated clusters; 0.2–0.3 weak separation; and near or below 0 no exploitable structure. Removing $\mathcal{L}_{\text{Local}}$ collapses the score below the initial-embedding level toward this structureless regime, matching the MRR drop of roughly an order of magnitude across all embedders (Table III). For nomic-embed-text and all-MiniLM-L6-v2, even with all losses active, the silhouette only recovers to its initial-embedding level: training cannot substantially improve their weak starting geometry. This is consistent with their low final MRR (Table III) and their diffuse embedding spaces (Fig. 8).



(a) All-minilm (b) Nomic-embed-text

Fig. 8. PCA projection for one table with three questions: raw LLM embeddings (left of each pair) versus UTUEL-refined embeddings (right), for a stronger embedder (all-MiniLM, MRR 0.674) and a weaker one (nomic-embed-text, MRR 0.220; Table III). The questions are: q_0 = *"What's Dorain Anneck's pick number?"*; q_1 = *"What is the nationality of the player from Vancouver Canucks?"*; q_2 = *"What's the pick number of the player from Springfield Olympics (Nejhl)?"*. The table embedding is computed *once* and reused for all three. In the raw space, no query lies close to its answer cell. After UTUEL, each target cell appears as multiple context-dependent copies, and for all-MiniLM each query (tracked by colour) is placed close to the copy matching its context. For nomic-embed-text, the query embeddings remain overlapping and none is placed near its target, indicating its initial geometry is too weakly structured for UTUEL to reorganise, which explains its poor retrieval. A full breakdown is in Section 9 of the Appendix.

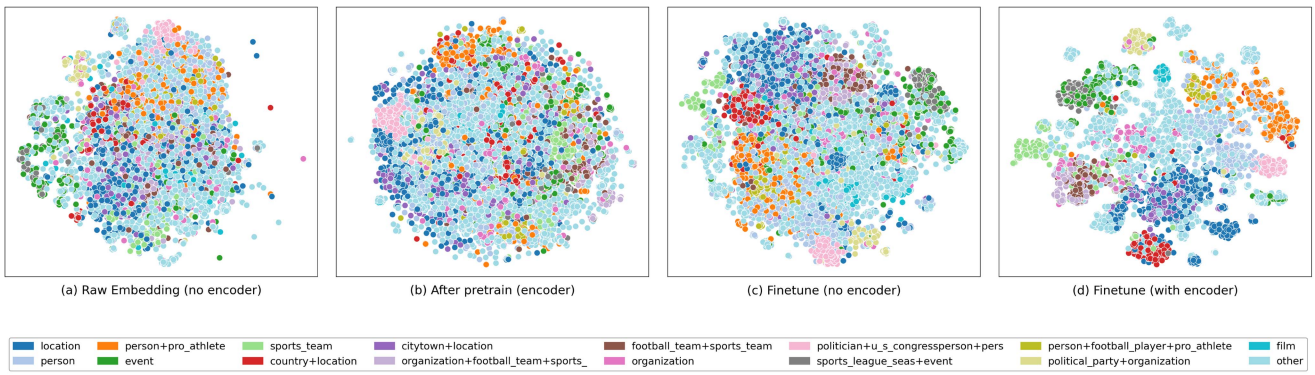


Fig. 9. Column-level embeddings for CTA on the TURL-CTA test split, mean-pooled per column and projected to 2D, coloured by ground-truth type. Left to right: raw, after UTUEL pretraining, fine-tuned without and with the encoder. Same-type columns grow more separable under UTUEL, showing the cell geometry transfers to column-level semantics.

additional tasks, such as CPA and Table Similarity Search, as well as on relational database tables and more diverse domains.

b) Dependence on the initial embedder: UTUEL refines an existing embedding space rather than creating one from scratch, so its gains are bounded by the quality of the initial embedder; for example, it does not improve nomic-embed-text. We have not yet characterised which embedder properties predict successful refinement, though we suspect the training objective and embedding type, such as symmetric versus asymmetric embeddings, play an important role. Training UTUEL from scratch with a learnable embedding layer remains unexplored.

c) Distribution shift: Training and test largely share domain and query style; robustness to unseen domains, layouts, or distant phrasings is only partially probed (see Fig. 5) and remains open.

d) Scope and accuracy: UTUEL currently supports single-cell answering and performs best on look-up queries, remaining weaker on complex ones (see Fig. 6). Compositional and aggregation queries are not yet addressed; a natural next step is to decompose them into traceable UTUEL look-ups, for example through chain-of-thought query decomposition, and to compare this with interpretable text-to-SQL baselines.

VI. Conclusion

The rise of LLMs has drawn attention away from how (semi-)structured data is represented, where progress still lags. A central reason, we argue, is the absence of a context-aware cell-level embedding for tables, comparable to word embeddings in NLP, that can jointly address semantic complexity, contextual variability, and structural heterogeneity. Processing tables as flat token sequences overlooks this structure and the relational semantics it carries. We addressed this with UTUEL by introducing *TRL-JEPA*, a Joint-Embedding Predictive Architecture in which an encoder learns context-aware cell embeddings and a predictor maps queries into the same learned space. Guided by the TRL principles of local alignment and global repulsion over Semantic Meta-Paths, UTUEL produces a single, reusable representation that supports Table Cell Retrieval, Table Question Answering, Column Type Annotation, and Table Retrieval. Its impact is threefold. First, embeddings are computed once per table and reused across queries and tasks at no further cost, avoiding the repeated per-query table re-encoding that prompted LLMs incur. Second, the learned cell-level geometry transfers across tasks without retraining, substantially improving retrieval over generic embedders with only a 2.9–11.8M-parameter encoder. Third, because answers are retrieved as specific table cells rather than generated text, they are grounded in the table and inherently mitigate hallucination. Together, these establish cell-level embeddings as a unified, low-cost foundation for table understanding, with potential to advance Intelligent Document Processing, agentic multimodal reasoning, Table RAG, and human-compatible database interaction.

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